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Vellore Institute of Technology
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School of Computer Science Engineering and Information Systems

Course Name: Soft Computing

Course Code: SWE1011

Faculty: Bharani Dharan N

Slot: A2 + TA2

Submitted in fulfilment for the Jth Component Project of SWE1011 - Soft Computing

Project Title: Household Energy Consumption Prediction

Project Final Report

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1. Introduction

With the rapid growth in electricity demand and increasing awareness of energy conservation, predicting household energy consumption has become an important area of research. Efficient energy usage not only helps reduce electricity costs for individuals but also supports sustainable resource management at a larger scale. However, energy consumption patterns in households are influenced by multiple dynamic factors such as appliance usage, weather conditions, and seasonal variations, making accurate prediction a challenging task.

This project presents a **Household Energy Consumption Prediction System**, a web-based application developed using Flask that integrates both machine learning and deep learning techniques to estimate energy usage. The system allows users to input appliance usage details along with environmental parameters like temperature, humidity, wind speed, and solar radiation. Based on these inputs, it predicts energy consumption through multiple models including Linear Regression, Random Forest, Gradient Boosting, and LSTM.

The application not only provides predictions but also enhances user understanding through interactive visualizations and real-time calculations. By combining data-driven modeling with an intuitive interface, the system aims to promote energy awareness and enable users to make informed decisions about their electricity consumption.

2. Problem Statement

Accurately predicting household energy consumption is complex due to the variability in user behavior, diversity of appliances, and the influence of changing weather conditions. Traditional estimation methods often fail to capture these dynamic patterns, leading to inefficient energy usage and lack of awareness among users.

There is a need for a system that can:

- Analyze multiple influencing factors such as appliance usage and environmental conditions
- Handle both linear and non-linear relationships in data
- Capture temporal patterns in energy consumption
- Provide accurate and user-friendly predictions

This project addresses these challenges by developing an intelligent prediction system that leverages machine learning and deep learning models to deliver reliable energy consumption forecasts, helping users optimize their energy usage and reduce unnecessary consumption.

3. Objectives

The main objective of this project is to design and develop an intelligent system that can accurately predict household energy consumption using advanced machine learning and deep learning techniques. The system focuses on combining user inputs, appliance usage patterns, and environmental factors to generate meaningful predictions. It also aims to provide an easy-to-use

interface that helps users understand their energy usage and make better decisions for efficient energy management.

Specific Objectives:

- To develop a web-based application using Flask for predicting household energy consumption
- To collect and utilize inputs such as appliance usage and weather conditions for accurate prediction
- To implement multiple machine learning models like Linear Regression, Random Forest, and Gradient Boosting for comparative analysis
- To integrate a deep learning model (LSTM) for capturing time-based energy consumption patterns
- To preprocess and normalize data using techniques like MinMaxScaler for improved model performance
- To provide real-time energy consumption calculations based on user inputs
- To visualize predictions using interactive charts for better understanding
- To classify weather conditions into seasons to enhance prediction accuracy
- To enable users to compare results from different models and choose suitable predictions
- To promote energy awareness and help users optimize electricity usage in households

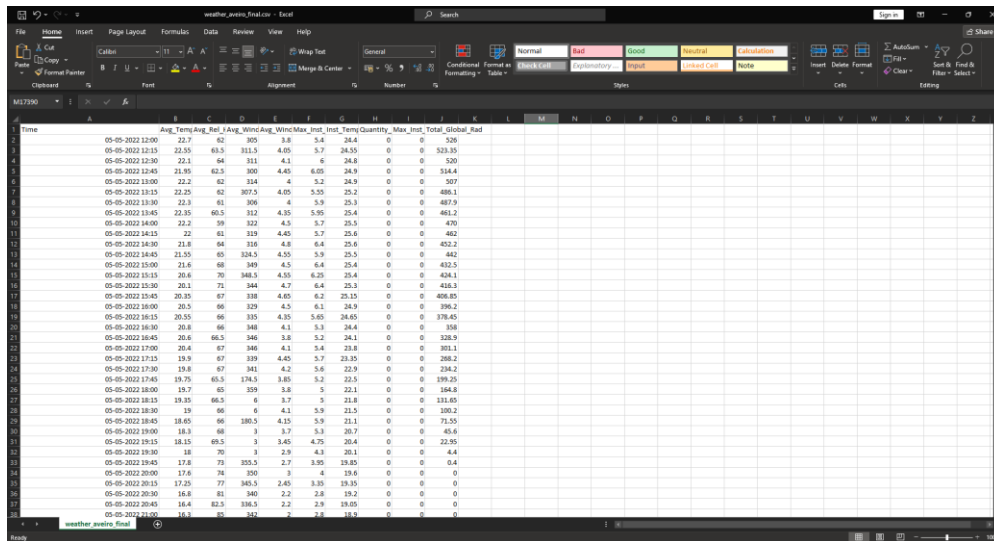
4. Dataset Description

- 1) Dataset Name : Electricity Consumption Dataset
- 2) Dataset Link : <https://data.mendeley.com/datasets/vryvyfz2tj/1>
- 3) Dataset File type : CSV
- 4) No. of Samples : Approximately 2000+ records (15-minute interval data)
- 5) No. of classes (if classification problem) : Not applicable (This is a Regression problem)

Energy / Electricity usage dataset :

[illegible]

Weather dataset :



Time	Avg_Temp	Avg_Rel_H	Avg_Wind	Avg_WindMax	Inst_Temp	Quantity	Max_Inst	Total_Global	Rad
05-05-2022 12:00	22.7	62	305	3.8	5.4	28.4	0	0	520
05-05-2022 12:15	22.55	63.3	311.3	4.05	5.7	24.55	0	0	523.35
05-05-2022 12:30	22.1	64	311	4.1	6	24.8	0	0	520
05-05-2022 12:45	21.95	63.5	300	4.45	6.05	24.9	0	0	514.4
05-05-2022 13:00	22.2	62	314	4	5.2	24.9	0	0	507
05-05-2022 13:15	22.25	62	307.5	4.05	5.55	25.2	0	0	486.1
05-05-2022 13:30	22.3	61	306	4	5.9	25.3	0	0	487.9
05-05-2022 13:45	22.35	60.5	312	4.35	5.95	25.4	0	0	461.2
05-05-2022 14:00	22.2	59	322	4.5	5.7	25.5	0	0	470
05-05-2022 14:15	22	61	319	4.45	5.7	25.6	0	0	462
05-05-2022 14:30	21.8	64	316	4.8	6.4	25.6	0	0	452.2
05-05-2022 14:45	21.55	65	324.5	4.55	5.9	25.5	0	0	442
05-05-2022 15:00	21.6	68	389	4.5	6.4	25.4	0	0	432.5
05-05-2022 15:15	20.6	70	348.5	4.55	6.25	25.4	0	0	424.1
05-05-2022 15:30	20.1	71	344	4.7	6.4	25.3	0	0	426.3
05-05-2022 15:45	20.35	67	338	4.85	6.2	25.15	0	0	406.85
05-05-2022 16:00	20.5	66	329	4.5	6.1	24.9	0	0	396.2
05-05-2022 16:15	20.55	66	335	4.55	5.65	24.65	0	0	379.45
05-05-2022 16:30	20.8	66	348	4.1	5.3	24.4	0	0	358
05-05-2022 16:45	20.6	66.5	346	3.8	5.2	24.1	0	0	328.9
05-05-2022 17:00	20.4	67	346	4.1	5.4	23.8	0	0	301.1
05-05-2022 17:15	19.9	67	339	4.45	5.7	23.35	0	0	286.2
05-05-2022 17:30	19.8	67	341	4.2	5.6	22.9	0	0	234.2
05-05-2022 17:45	19.75	65.5	174.5	3.85	5.2	22.5	0	0	199.25
05-05-2022 18:00	19.7	65	359	3.8	5	22.1	0	0	164.8
05-05-2022 18:15	19.35	66.5	6	3.7	5	21.8	0	0	111.65
05-05-2022 18:30	19	66	6	4.1	5.9	21.5	0	0	100.2
05-05-2022 18:45	18.65	66	180.5	4.15	5.9	21.1	0	0	71.55
05-05-2022 19:00	18.3	68	3	3.7	5.3	20.7	0	0	45.6
05-05-2022 19:15	18.15	68.5	3	3.45	4.75	20.4	0	0	22.55
05-05-2022 19:30	18	70	3	2.9	4.3	20.1	0	0	4.4
05-05-2022 19:45	17.8	73	255.5	2.7	3.95	19.85	0	0	0.4
05-05-2022 20:00	17.6	76	300	3	4	19.6	0	0	0
05-05-2022 20:15	17.25	77	345.5	2.45	3.35	19.35	0	0	0
05-05-2022 20:30	16.8	81	340	2.2	2.8	19.2	0	0	0
05-05-2022 20:45	16.4	82.5	336.5	2.2	2.9	19.05	0	0	0
05-05-2022 21:00	16.3	85	322	2	2.8	18.5	0	0	0

5. Fundamentals of ML / DL Techniques used

This project is built on core concepts of machine learning and deep learning, which enable the system to learn patterns from data and make accurate predictions. These techniques help in understanding the relationship between appliance usage, weather conditions, and energy consumption.

Machine Learning Fundamentals

Machine learning models learn from historical data by identifying patterns and relationships between input features (like temperature, appliance usage) and output (energy consumption). These models are trained using labeled datasets and are effective for both simple and complex prediction tasks.

Techniques Used:

- **Linear Regression**
A basic supervised learning algorithm that assumes a linear relationship between input variables and energy consumption. It is simple, fast, and provides easily interpretable results.
- **Random Forest**
An ensemble learning method that combines multiple decision trees to improve prediction accuracy and reduce overfitting. It works well with non-linear data and handles multiple features effectively.
- **Gradient Boosting**
Another ensemble technique where models are built sequentially, each correcting the errors of the previous one. It provides high accuracy and is robust for complex datasets.

Deep Learning Fundamentals

Deep learning models are inspired by the human brain and are capable of learning complex patterns, especially in sequential or time-based data. These models use neural networks with multiple layers to process information.

Technique Used:

- **LSTM (Long Short-Term Memory)**

A type of Recurrent Neural Network (RNN) designed to handle sequential data. It can remember past information over long periods, making it suitable for predicting energy consumption trends based on historical and time-dependent data such as weather changes and usage patterns.

Key Concepts Applied

- **Supervised Learning:** Models are trained using input-output pairs
- **Feature Engineering:** Selecting relevant inputs like appliances and weather data
- **Data Preprocessing:** Scaling data using MinMaxScaler for better performance
- **Model Training & Testing:** Evaluating models using performance metrics
- **Prediction & Visualization:** Displaying results in an interactive and understandable format

6. Methodology

The proposed Household Energy Consumption Prediction System follows a structured and systematic approach to ensure accurate and efficient prediction of energy usage. The methodology combines data collection, preprocessing, model training, and visualization in a step-by-step workflow, enabling the system to transform raw inputs into meaningful predictions.

The process begins with collecting user inputs related to household appliances and environmental conditions. These inputs are then processed and transformed into a suitable format for model prediction. Multiple machine learning and deep learning models are applied to analyze the data, and the results are presented through interactive visualizations for better understanding.

Step by Step Methodology :

1. Data Collection

The system collects historical data from CSV files containing household energy consumption and weather parameters. Additionally, real-time user inputs are gathered through the web interface, including appliance usage (in hours) and environmental factors such as temperature, humidity, wind speed, and solar radiation. This combined dataset serves as the input for prediction.

2. Input Processing

Energy consumption for each appliance is calculated using predefined power ratings. The computation is performed using the following formula:

$$\text{Energy (kWh)} = \frac{\text{Power (W)} \times \text{Time (hours)}}{1000}$$

The calculated values for all appliances are summed and combined with weather inputs to form a structured input dataset.

3. Season Classification

The system classifies the season automatically based on temperature values:

$$\begin{cases} \text{Hot} & : T > 30^{\circ}C \\ \text{Moderate} & : 20^{\circ}C \leq T \leq 30^{\circ}C \\ \text{Cold} & : T < 20^{\circ}C \end{cases}$$

This classification helps the model understand seasonal variations in energy consumption.

4. Data Preprocessing

The collected data is cleaned to handle missing or inconsistent values. Then, normalization is applied using Min-Max Scaling:

$$X_{scaled} = \frac{X - X_{min}}{X_{max} - X_{min}}$$

This ensures that all features are within a uniform range, improving model performance.

5. Feature Engineering

Relevant features such as appliance-wise energy usage, total consumption, weather conditions, and season category are selected and structured. This step enhances the model's ability to learn meaningful relationships between inputs and output.

6. Model Selection and Training

Multiple models are used and trained on historical data:

- Linear Regression (for baseline prediction)
- Random Forest (for handling non-linear relationships)
- Gradient Boosting (for improved accuracy)

- LSTM (for time-series and sequential patterns)

Each model learns the mapping between input features and energy consumption.

7. Prediction Generation

The trained models take processed user inputs and generate predicted energy consumption values. The system may provide outputs from multiple models to allow comparison and better decision-making.

8. Visualization of Results

Predicted results are displayed using interactive visualizations such as:

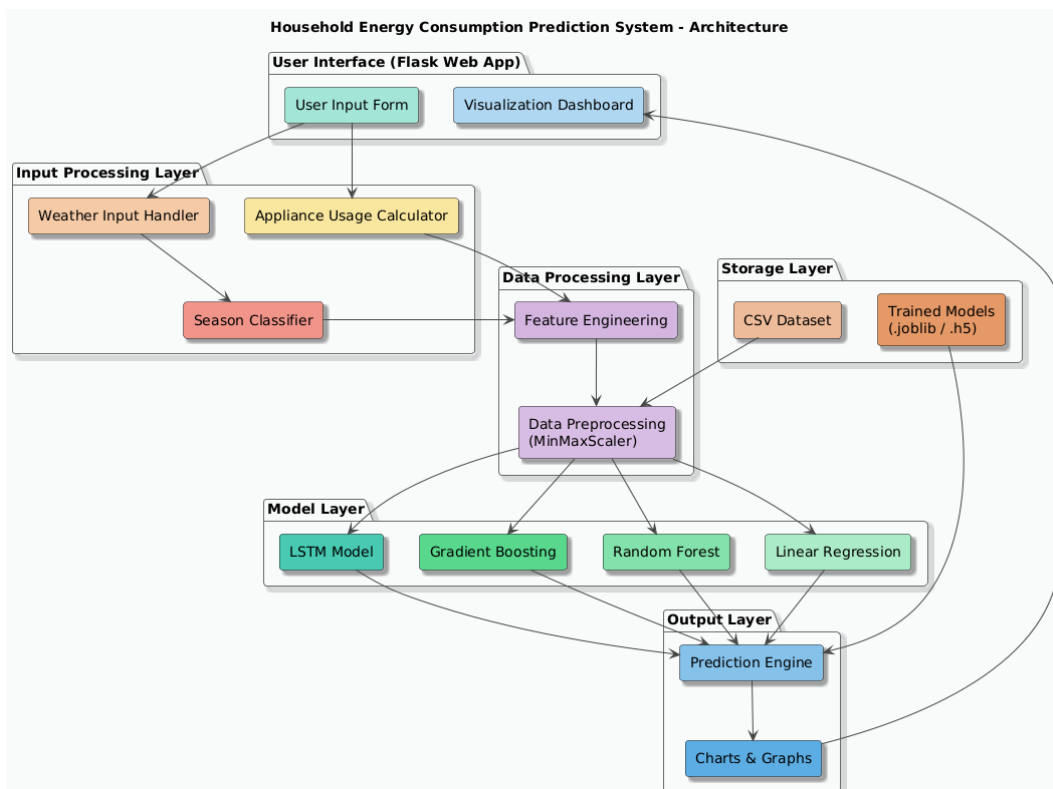
- Doughnut charts (appliance-wise energy distribution)
- Bar/line graphs (comparison of predicted vs total consumption)

These visuals make it easier for users to interpret the results.

9. System Deployment

The complete system is deployed using Flask as a web application. It provides a clean and user-friendly interface where users can input data, select models, and view predictions instantly, ensuring accessibility and practical usability.

Flowchart / Architecture Diagram :



Architecture Explanation

The architecture of the Household Energy Consumption Prediction System is organized into multiple layers that work together to process user inputs, perform predictions, and display results. This layered design ensures clarity, modularity, and efficient data flow throughout the system.

1. User Interface Layer

This is the front-end layer where users interact with the system. It includes the User Input Form, where users provide appliance usage details and weather parameters such as temperature, humidity, wind speed, and solar radiation. The Visualization Dashboard presents the prediction results using charts and graphs, helping users easily understand their energy consumption.

2. Input Processing Layer

In this layer, the raw inputs from the user are processed. The Appliance Usage Calculator computes the energy consumed by each appliance based on its power rating and usage duration. The Weather Input Handler manages environmental data provided by the user. The Season Classifier then determines whether the condition is hot, moderate, or cold based on temperature. All these components work together to prepare meaningful input data for further processing.

3. Data Processing Layer

This layer transforms the processed inputs into a suitable format for model prediction. Feature Engineering organizes and selects important features such as appliance energy, weather parameters, and season category. Data Preprocessing, using techniques like MinMaxScaler, normalizes the data so that all features are on a similar scale, improving model efficiency and accuracy.

4. Storage Layer

The storage layer holds both the datasets and trained models. CSV datasets provide historical energy and weather data used during training. The trained models, saved in formats like .joblib and .h5, are loaded when needed for prediction. This layer ensures that data and models are easily accessible without retraining every time.

5. Model Layer

This layer consists of multiple prediction models including Linear Regression, Random Forest, Gradient Boosting, and LSTM. Each model analyzes the processed input data and learns patterns to estimate energy consumption. The presence of multiple models allows comparison and improves the flexibility of the system.

6. Output Layer

The final layer generates and presents the results. The Prediction Engine receives outputs from different models and produces the final energy consumption value. The Charts and Graphs component visualizes the results, showing appliance-wise energy distribution and overall consumption, which helps users interpret the predictions clearly.

Novelty :

- Uses real-world smart meter and weather data together.
- Compares Machine Learning and Deep Learning models trained from scratch.
- Focuses on numerical time-series energy prediction rather than images or text.

7. Results :

Tabulate the performance metrics of each model :

Model	MAE	MSE	RMSE	R ²
Random Forest	0.8054	1.2973	1.1390	0.9170
Linear Regression	1.8678	5.9548	2.4403	0.6190
Gradient Boosting	1.2666	2.8542	1.6894	0.8174
LSTM	0.8813	1.4687	1.2119	0.9060

Key Observations

- Random Forest achieves the lowest error values (MAE, MSE, RMSE) and the highest R² score, indicating the most accurate and reliable predictions.
- LSTM performs closely to Random Forest with slightly higher error values and a marginally lower R² score, showing strong predictive capability.
- Gradient Boosting provides moderate performance with higher errors and a lower R² compared to the top two models.
- Linear Regression shows the weakest performance with the highest error values and the lowest R² score, indicating poor model fit.

Highest Achieved Model

Random Forest achieved the best performance among all models.

It has the highest R² score (0.9170), which means it explains about **91.7%** of the variance in the data, indicating a very strong fit. At the same time, it records the lowest MAE (0.8054), MSE (1.2973), and RMSE (1.1390), showing that its predictions have the smallest errors compared to the other models.

This combination of high explanatory power and low prediction **error** makes Random Forest the most accurate and reliable model for this dataset.

8. Conclusion

The Household Energy Consumption Prediction System successfully demonstrates how machine learning and deep learning techniques can be applied to solve real-world problems related to energy management. By integrating multiple models such as Linear Regression, Random Forest, Gradient Boosting, and LSTM, the system is able to analyze both simple and complex patterns in data and provide reliable predictions.

The project effectively combines user inputs, appliance usage details, and weather conditions to generate meaningful insights into energy consumption. The inclusion of real-time calculations, season-based analysis, and interactive visualizations enhances user understanding and makes the system practical for everyday use.

Overall, this system not only improves prediction accuracy but also promotes awareness about energy usage, helping users make informed decisions to optimize their consumption. It serves as a scalable and efficient solution that can be further extended for smart home systems and sustainable energy management applications in the future.

9. References

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